

Audience Sensitivity, Content Compliance & Format Preference Analysis of the France Top 50 Playlist

*A Data-Driven Study of Chart Representation, Release Format, Duration, and
Album Structure for Atlantic Recording Corporation*

Abstract

Prepared for: Atlantic Recording Corporation

Dataset: Daily France Top 50 Playlist Snapshot Data, 18 May 2024 – 27 November 2025

Document type: Academic research paper with executive summary

By

Paris Asegaonkar

Table of Contents

Sr. No	Topic	Page No.
	Abstract	<u>1</u>
1.	Introduction	<u>2</u>
2.	Industry and Business Context	<u>3</u>
3.	Problem Statement	<u>3</u>
4.	Research Objectives	<u>4</u>
5.	Research Questions	<u>4</u>
6.	Scope of the Study	<u>5</u>
7.	Dataset Description	<u>5</u>
8.	Data Validation & Cleaning	<u>6</u>
9.	Methodology	<u>7</u>
10.	Feature Engineering	<u>7</u>
11.	KPI Framework	<u>8</u>
12.	Exploratory Data Analysis	
	12.1. Overall Dataset Profile	<u>9</u>
	12.2. Explicit vs Clean Content	<u>9</u>
	12.3. Release Format	<u>9</u>
	12.4. Album Structure	<u>10</u>
	12.5. Song Duration	<u>10</u>
	12.6. Rank Tire Comparison	<u>10</u>
	12.7. Monthly Trends	<u>15</u>
13.	Statistical Analysis	<u>15</u>
14.	Content Acceptance Score Analysis	<u>16</u>
15.	Key Findings	<u>16</u>
16.	Discussion	
	16.1. Explicit-Content Strategy	<u>17</u>
	16.2. Clean & Explicit Version Planning	<u>17</u>
	16.3. Playlist Pitching	<u>18</u>
	16.4. Release Format Planning	<u>18</u>
	16.5. Album Tracklist Strategy	<u>18</u>
	16.6. Song Length Considerations	<u>18</u>
	16.7. France Specific Localization	<u>18</u>
	16.8. Monitoring of Changes Over Time	<u>18</u>
17.	Strategic Implications for Atlantic Recording Corporation	<u>18</u>
18.	Recommendations	<u>19</u>
19.	Limitations	<u>20</u>
20.	Conclusion	<u>21</u>
	References	<u>22</u>
	Appendix	<u>22</u>

Abstract

This study examines the structural characteristics of the France Top 50 playlist using a cleaned daily snapshot dataset covering 555 days between 18 May 2024 and 27 November 2025 (27,749 analytical records). The analysis was commissioned by Atlantic Recording Corporation to inform France-specific decisions on content sensitivity, explicit versus clean release planning, release format, song duration, and album tracklist architecture. Because the dataset consists of repeated daily chart snapshots rather than individual-level listener records, all findings are interpreted strictly as patterns of chart representation and observed association, not as causal effects or universal statements about French listener preference.

Key results show that explicit content accounts for 56.27% of all analytical tracks and rises to 62.44% within the Top 10 tier, indicating that explicit content is not absent from the French chart but becomes more concentrated toward the highest ranks. At the same time, clean tracks record a higher project-defined Content Acceptance Score (65.30 versus 62.67), producing a nuanced dual finding. The full Top 50 is slightly album-heavy (52.89% album tracks versus 47.11% singles), while singles are marginally more represented in the Top 10 (50.75%) and Top 25 (50.99%). Average song duration is exceptionally stable across rank tiers (approximately 3.09 minutes), and the 3–4-minute duration bucket shows the strongest average chart position (24.89). Tracks associated with smaller album tracklists (1–10 tracks) show the strongest average popularity (77.04) and rank (25.64), and the project-defined Album Size Impact Index stands at 4.67. Monthly trends reveal material variation in explicit-content representation, from a high of 67.03% (October 2024) to a low of 45.27% (July 2025), while duration remains comparatively stable. Exploratory Welch t-tests indicate statistically significant group differences ($p < 0.001$), though repeated daily observations violate independence assumptions and the tests are treated as exploratory only.

The paper translates these findings into a four-pillar "France Top 50 Content Strategy Framework" covering content sensitivity, release format, structural optimization, and continuous monitoring, and concludes that the data supports a dual-version, evidence-based localization strategy rather than any blanket clean-only, explicit-only, single-only, or album-only rule.

Keywords: France Top 50; playlist analytics; explicit content; clean content; release format; album structure; song duration; chart representation; content localization; Atlantic Recording Corporation

1. Introduction

Playlist charts have become one of the most visible arenas in which recorded music competes for attention, and national charts such as the France Top 50 provide a structured, repeatedly observable window into which songs achieve sustained chart representation in a given market. For a global label such as Atlantic Recording Corporation, decisions about which versions of a song to release, whether to prioritize singles or album tracks, how long songs should be, and how many tracks an album should contain are frequently made on the basis of global heuristics. Yet chart structure can differ meaningfully across national markets, and a chart-level profile specific to France offers an evidence base for localizing release and pitching strategy.

This study analyzes a daily snapshot dataset of the France Top 50 playlist over an approximately eighteen-month observation window (18 May 2024 to 27 November 2025). The study addresses three intersecting business questions. First, a content-sensitivity question: how is explicit content represented on the French chart, and does explicit status appear to be associated with differences in chart performance? Second, a format question: are singles or album tracks more strongly represented overall, and does the answer change between the Top 10, the Top 25, and the full Top 50? Third, a structural question: how are song duration and album tracklist size associated with popularity and chart position?

The analysis is deliberately framed in terms of chart representation. The dataset contains no individual listeners, no streaming counts by user, no skip or completion behavior, and no survey responses. It therefore cannot establish what French listeners prefer in the behavioral sense, nor can it establish that any content attribute causes stronger chart performance. What it can do and what this paper does is characterize the attributes of tracks that achieve stronger representation and higher positions in the observed France Top 50 data, quantify those patterns, test them for robustness, and translate them into carefully bounded strategic implications.

The remainder of the paper is organized as follows. Sections 2–6 establish the industry context, problem statement, objectives, research questions, and scope. Sections 7–11 describe the dataset, the validation and cleaning process, the methodology, the derived features, and the KPI framework. Section 12 presents the exploratory data analysis. Sections 13–14 report the exploratory statistical tests and the Content Acceptance Score analysis. Sections 15–18 present key findings, discussion, strategic implications, and recommendations. Sections 19–20 set out limitations and conclusions, followed by references and an appendix containing KPI formulas and supporting statistical detail.

2. Industry and Business Context

Recorded-music labels operating across multiple territories face a recurring portfolio problem: a single global release strategy must be adapted to local market structures without fragmenting the underlying catalog investment. Decisions that appear minor at the level of an individual release whether to ship an explicit and a clean version, whether to service a single or an album cut to playlist editors, how many tracks to include on a standard album compound across a release slate into material commercial differences.

Playlist pitching is a central element of this problem. Editorial playlist placement is influenced by factors that include, but are not limited to, track quality, artist momentum, genre fit, and the suitability of a given recording for the playlist context. Explicit-content labeling interacts with this process: some playlist contexts admit explicit recordings, others require clean versions, and some markets exhibit different tolerances than others. A label that understands how explicit and clean recordings are actually represented on a target market's flagship chart is better positioned to plan dual-version releases, timing, and pitching sequences.

Release architecture raises a parallel set of questions. The single-versus-album balance determines how catalog is sequenced and how playlist opportunities are allocated across tracks. Album tracklist length affects how attention is distributed across a body of work and, plausibly, how any individual track performs although such an effect, if it exists, is entangled with artist popularity, genre, and release strategy. Song duration interacts with playlist programming economics and editorial preferences, and the industry has repeatedly debated the consequences of changing track lengths for streaming-era consumption.

For Atlantic Recording Corporation, France represents a major market in which these decisions must be made release by release. This study provides a France-specific, chart-level evidence base for those decisions. Importantly, the study makes no claim about French cultural preferences beyond what the data supports; its contribution is empirical description of chart structure and a disciplined translation of that description into strategy options.

3. Problem Statement

Atlantic Recording Corporation requires an evidence base for France-specific release, content-localization, playlist-pitching, and compliance decisions, but lacks a structured analysis of how explicit content, release format, song duration, and album structure are associated with chart representation on the France Top 50 playlist. Without such an analysis, decisions risk being driven by global heuristics that may not reflect the observed structure of the French chart, or by oversimplified assumptions for example, that the French market is uniformly "clean" in its chart content, or that one release format dominates at all chart levels.

The problem is compounded by data considerations. The available data is a repeated daily snapshot of chart positions and track metadata, not listener-level behavioral data. Any analysis must therefore (a) extract maximum structured insight from chart-level observations, (b) avoid over-interpreting those observations as listener preference or causal effect, and (c) handle data-quality issues including a duplicated daily snapshot and an incomplete record transparently. This paper addresses the problem statement under those constraints.

4. Research Objectives

The study pursues nine objectives:

- (1) To measure the representation of explicit content across the France Top 50 and within higher-ranking tiers.
- (2) To compare the chart performance of explicit and clean tracks using popularity, chart position, and a project-defined composite acceptance score.
- (3) To evaluate the relative representation of single releases and album tracks across the Top 10, Top 25, and Top 50.
- (4) To analyze the association between album tracklist size and track-level popularity and chart position.
- (5) To evaluate duration patterns and identify the duration ranges most strongly represented among higher-ranking tracks.
- (6) To compare content-attribute distributions across Top 10, Top 25, and Top 50 rank tiers.
- (7) To construct and interpret a project-defined Content Acceptance Score that combines chart position and popularity.
- (8) To synthesize the preceding results into a data-supported France Top 50 content profile.
- (9) To translate that profile into strategic recommendations for Atlantic Recording Corporation covering release planning, version management, playlist pitching, and monitoring.

5. Research Questions

- RQ1.** How does explicit-content representation differ across the France Top 50 as a whole and within higher-ranking tiers (Top 10, Top 25)?
- RQ2.** How is explicit versus clean content associated with popularity and chart position?
- RQ3.** How are singles and album tracks distributed across the Top 10, Top 25, and Top 50 tiers?
- RQ4.** How is album size associated with track popularity and chart position?
- RQ5.** Which duration ranges are most strongly represented among higher-ranking tracks?
- RQ6.** Which combination of content attributes characterizes stronger France Top 50 representation?
- RQ7.** What strategic implications do these observed patterns provide for Atlantic Recording Corporation?

Consistent with the observational nature of the data, the study evaluates five exploratory hypotheses rather than confirmatory hypotheses:

- H1.** Explicit status is associated with differences in chart performance (popularity and position).
- H2.** Single and album tracks differ in average popularity.
- H3.** Album size is associated with track popularity.
- H4.** Song duration is associated with chart position and popularity.
- H5.** The distribution of content attributes differs between the Top 10, Top 25, and Top 50 tiers.

These hypotheses are labeled exploratory because the dataset is observational, contains repeated observations of the same tracks, and offers no mechanism for causal identification.

6. Scope of the Study

The study is bounded in four dimensions. Temporally, it covers daily snapshots from 18 May 2024 through 27 November 2025 (555 dates). Structurally, it covers the 50 chart positions of the France Top 50 playlist only; no other French charts, no global charts, and no other playlists are included. Analytically, it examines track-level attributes available in the dataset: explicit flag, album type, duration, album size (total tracks), popularity score, and chart position. No genre, demographic, streaming-volume, or behavioral variables are available, and none are analyzed. Interpretively, all conclusions are statements about France Top 50 chart representation and observed association; the study does not claim to measure the preferences of all French listeners, nor does it claim causal effects of any attribute on chart performance.

Compilation records are retained in the overall dataset for completeness but are excluded from binary single-versus-album comparisons and from the KPI values and format charts displayed in the project dashboard, which uses the filter Album Type = album + single. This treatment is applied consistently throughout the paper and is revisited in Sections 8 and 19.

7. Dataset Description

The raw dataset contained approximately 27,800 rows, structured as 50 playlist positions per day across 555 daily snapshots. Each record represents one track occupying one chart position on one date, and carries the fields described in Table 1. The same song may appear on many consecutive dates; the unit of observation is therefore the track-date record, not the unique song.

Table 1. Dataset fields and descriptions

Field	Description	Role in analysis
date	Calendar date of the daily chart snapshot	Time indexing; monthly trend analysis
position	Chart position on the given date (1–50)	Rank-tier classification; rank scoring
song	Track title	Identification; record completeness checks
artist	Performing artist name	Identification (no artist-level modeling performed)
popularity	API-derived popularity score (0–100)	Performance KPI input; group comparisons
duration (duration_ms)	Track length in milliseconds; converted to minutes (duration_min)	Duration analysis and bucketing
album type	Release container type: single, album, or compilation	Format analysis
total tracks	Number of tracks on the associated album	Album-size analysis and bucketing
explicit flag	Binary explicit-content indicator	Content-sensitivity analysis
album cover URL	Link to album artwork	Dashboard display only; not analyzed

The final cleaned analytical dataset comprises 27,749 records: 555 dates, 50 positions per day except one date (which carries 49 records after removal of an incomplete record), 15,614 explicit records, 12,135 clean records, 14,672 album records, 13,068 single records, and 9 compilation records.

8. Data Validation and Cleaning

Data validation proceeded in five stages. First, date coverage was verified: the data spans 18 May 2024 through 27 November 2025, comprising 555 daily snapshots. Second, position integrity was checked: positions should fall within 1–50 for every date. Third, daily record counts were validated against the intended structure of 50 records per date. Fourth, duplicate date-position combinations were screened. Fifth, field-level completeness was assessed, leading to the removal of one record with a missing song title.

The validation surfaced one major anomaly. On 1 March 2025, the raw data contained 100 records rather than the intended 50, indicating a duplicated daily snapshot. To preserve the design principle that every date receives equal weight, the cleaning workflow retained the first 50-row snapshot for that date and removed the duplicate. After this correction, 27,750 records remained. One further record with a missing song title was removed, yielding the final analytical dataset of 27,749 records. Consequently, one date carries 49 records; the cleaned dataset is not perfectly complete, and this residual imperfection is disclosed rather than concealed.

Two standardization steps completed the cleaning: duration values were converted from milliseconds to minutes ($\text{duration_min} = \text{duration_ms} / 60,000$), and the album-type and explicit-flag fields were standardized to consistent categorical values (single / album / compilation; explicit / clean). The nine compilation records were retained in the overall dataset but flagged for exclusion from binary format comparisons. Table 2 summarizes the validation and cleaning workflow.

Table 2. Data validation and cleaning summary

Step	Issue identified	Action taken	Records affected
Date coverage check	18 May 2024 – 27 November 2025; 555 dates	Confirmed as expected structure	—
Daily record-count check	1 March 2025 contained 100 records (duplicated snapshot)	Retained first 50-row snapshot for that date so every date has equal weight	50 records removed
Completeness check	One record with missing song title	Record removed	1 record removed
Post-cleaning dataset	27,749 records	Final analytical dataset; one date has 49 records	27,800 → 27,749
Field standardization	Mixed units and categorical spellings	Duration converted to minutes; album type and explicit flag standardized	All records
Compilation handling	9 compilation records in raw mix	Retained overall; excluded from binary single-vs-album analysis and dashboard KPI filter	9 records flagged

These decisions matter for two reasons. The duplicated-snapshot correction prevents double-counting of one day, which would otherwise bias every daily-average and share-based KPI computed on the affected date. The incomplete-record removal trades a trivial loss of coverage (one position on one date) for categorical integrity in title-based identification. Neither correction is presented as making the dataset perfect; both are documented so that downstream users can assess residual risk.

9. Methodology

The analytical workflow combines six techniques, in a deliberately sequenced progression from description to cautious inference.

(1) Descriptive statistics. Shares, means, and counts were computed for explicit content, release format, duration, and album size, overall and by subgroup. All headline KPIs (Section 11) are descriptive.

(2) Group comparison. Explicit versus clean tracks and single versus album tracks were compared on mean popularity, mean chart position, and the composite Content Acceptance Score.

(3) Rank-tier analysis. The chart was segmented into Top 10 (positions 1–10), Top 25 (positions 1–25), and Top 50 (positions 1–50), and content-attribute profiles were computed within each tier. Because the tiers are nested, tier differences describe where on the chart a given attribute is concentrated; they do not compare independent samples.

(4) Correlation analysis. Bivariate associations among duration, album size, popularity, and position were examined as exploratory correlation patterns, reported in Section 12 alongside the composite findings and summarized for the correlation-matrix exhibit (Figure 16).

(5) Exploratory significance testing. Welch two-sample t-tests (which do not assume equal variances) were applied to three comparisons: explicit versus clean popularity, explicit versus clean chart position, and single versus album popularity. The resulting p-values are reported as exploratory significance testing of observed group differences. Because the same songs recur across daily snapshots, observations are not independent; the tests therefore overstate effective sample size and are not treated as confirmatory evidence.

(6) Composite KPI development and monthly trend analysis. A project-defined Content Acceptance Score combining rank and popularity was constructed (Section 11), and monthly aggregates were computed to assess stability over time.

Throughout, three epistemic levels are kept distinct: descriptive analysis (what the data shows), exploratory statistical analysis (whether observed differences are large relative to sampling noise under imperfect assumptions), and business interpretation (what Atlantic might prudently do given the observed pattern). Findings are only ever promoted from one level to the next with explicit caveats

10. Feature Engineering

Six derived features were constructed from the cleaned fields:

Table A (feature engineering). Derived analytical features

Feature	Definition	Analytical purpose
duration_min	duration_ms converted to minutes	Duration analysis in interpretable units
rank_tier	Top 10 = positions 1–10; Top 25 = positions 1–25; Top 50 = positions 1–50	Nested tier profiling
rank_group	1–10; 11–25; 26–50	Non-nested band comparisons
duration_bucket	<2 min; 2–3 min; 3–4 min; >4 min	Duration pattern analysis
album_size_bucket	Small (1–10); Medium (11–15); Large (16–20); Very Large (21+)	Album-structure analysis
format_group	Single; Album; Compilation	Release-format analysis

The bucket boundaries reflect conventional industry segmentations (three-to-four-minute songs; standard album lengths) and were fixed before analysis to avoid data-driven boundary selection. The rank_tier and rank_group features encode the same positional information at different granularities; both are reported where the tier view (nested) and the band view (non-nested) answer different questions.

11. KPI Framework

Eight KPIs were defined. Formulas are stated exactly, and project-defined metrics are labeled as such. Where the dashboard applies the Album Type = album + single filter (excluding the nine compilation records), this is noted.

1. Explicit Content Share = explicit tracks ÷ all analytical tracks × 100. Overall value: 56.27%.
2. Clean Content Share = clean tracks ÷ all analytical tracks × 100. Overall value: 43.73%.
3. Clean Dominance Ratio = clean tracks ÷ explicit tracks. Overall value: 0.78. This is a project-defined ratio, not an industry-standard KPI; a value below 1.0 indicates that clean records are fewer than explicit records in the dataset.
4. Single vs Album Ratio = single tracks ÷ album tracks. Overall value: 0.89. Compilation records are excluded from this binary comparison.
5. Average Song Duration = mean duration in minutes. Overall value: 3.09 minutes.
6. Album Size Impact Index (project-defined) = mean popularity of tracks on large albums (>10 tracks) – mean popularity of tracks on small albums (≤10 tracks). Overall value: –4.67. Interpretation: under this project definition, tracks associated with larger albums averaged 4.67 popularity points lower than tracks associated with smaller albums.
7. Average Popularity = mean API-derived popularity score. Overall value: 76.64 (0–100 scale).
8. Content Acceptance Score (project-defined composite indicator, not an industry-standard measure). First compute Rank Score = $100 \times (51 - \text{position}) \div 50$, which maps position 1 to 100 and position 50 to 2. Then Content Acceptance Score = $0.5 \times \text{Rank Score} + 0.5 \times \text{Popularity}$, yielding a 0–100 composite. Overall value: 63.82. Component values: clean tracks average 65.295179 (≈65.30); explicit tracks average 62.672826 (≈62.67).

Table 3. Overall KPI summary

KPI	Formula (abbreviated)	Overall value	Type
Explicit Content Share	Explicit ÷ all tracks × 100	56.27%	Standard share
Clean Content Share	Clean ÷ all tracks × 100	43.73%	Standard share
Clean Dominance Ratio	Clean ÷ explicit	0.78	Project-defined
Single vs Album Ratio	Singles ÷ albums	0.89	Standard ratio (binary scope)
Average Song Duration	Mean duration_min	3.09 min	Standard mean
Average Popularity	Mean popularity score	76.64	API-derived mean
Album Size Impact Index	Large-album pop. – small-album pop.	–4.67	Project-defined
Content Acceptance Score	$0.5 \times \text{Rank Score} + 0.5 \times \text{Popularity}$	63.82	Project-defined composite

The composite score weights chart position and popularity equally. This is an analytical convenience, not a claim about their equal commercial importance; sensitivity of conclusions to the 50/50 weighting was considered qualitatively, and the direction of the explicit-clean gap (Section 14) is consistent with the component-level patterns.

12. Exploratory Data Analysis

12.1 Overall Dataset Profile

The analytical dataset comprises 27,749 track-date records across 555 days. Average popularity across all records is 76.64 on the 0–100 API-derived scale; average duration is 3.09 minutes. The content mix is majority explicit (56.27%), and the format mix is majority album tracks (14,672 album records versus 13,068 single records; the nine compilation records are negligible in volume and excluded from binary format views). Figure 1 (project analytical framework) situates these descriptive profiles within the workflow from data validation through KPI computation to strategic translation; Figures 2 and 4 display the explicit-clean and single-album marginal distributions respectively.

12.2 Explicit vs Clean Content

Explicit tracks account for 15,614 records (56.27%) and clean tracks 12,135 records (43.73%); the Clean Dominance Ratio of 0.78 confirms the explicit majority. The dashboard exhibits below (from the supplied Top_50.pdf dashboard screenshot, Overview tab, with Album Type = album + single) visualize this split and its tier variation. The salient pattern is tier gradient: explicit share rises from 56.27% across the full Top 50 to 57.70% in the Top 25 and 62.44% in the Top 10 (Figure 3). In other words, explicit content is not merely tolerated on the French chart; it is disproportionately concentrated at its highest ranks.

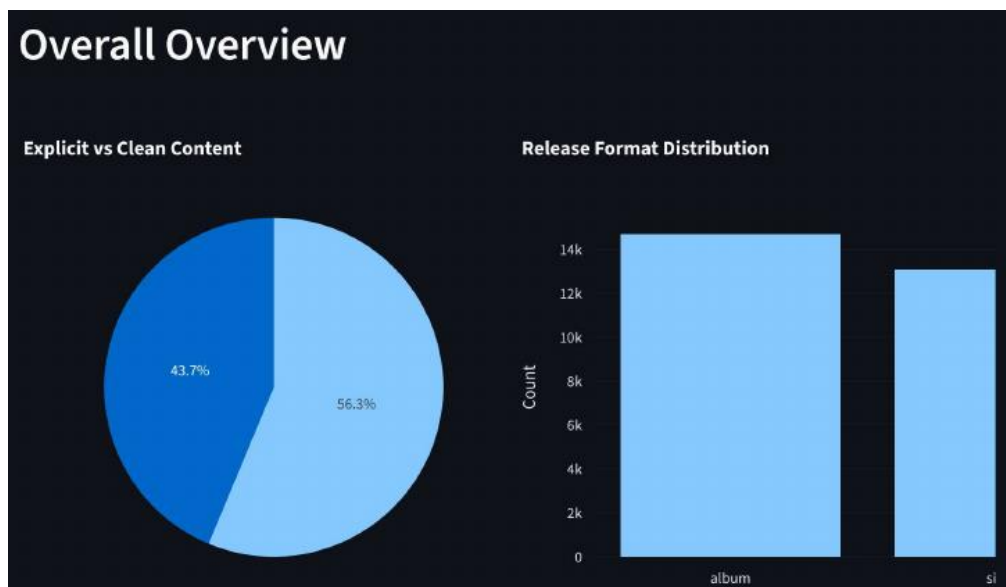


Exhibit A — Dashboard view (Top 50 filter, Album Type = album + single): Explicit vs Clean content pie (56.3% / 43.7%) and Release Form Distribution bars. Source: Top_50.pdf, Overview tab.

12.3 Release Format

Across the full Top 50, album tracks outnumber single tracks (52.89% versus 47.11%; Single vs Album Ratio 0.89). The balance reverses modestly at the top: singles constitute 50.75% of the Top 10 and 50.99% of the Top 25. Singles are therefore slightly more concentrated in higher ranks even though albums dominate the full chart (Figures 4–6). The second dashboard exhibit below (Top_25.pdf) illustrates the near-parity of the format split within a higher tier. This

pattern is descriptive of representation, not evidence that French consumers prefer one format; playlist editorial behavior and release scheduling both shape which format occupies which positions.

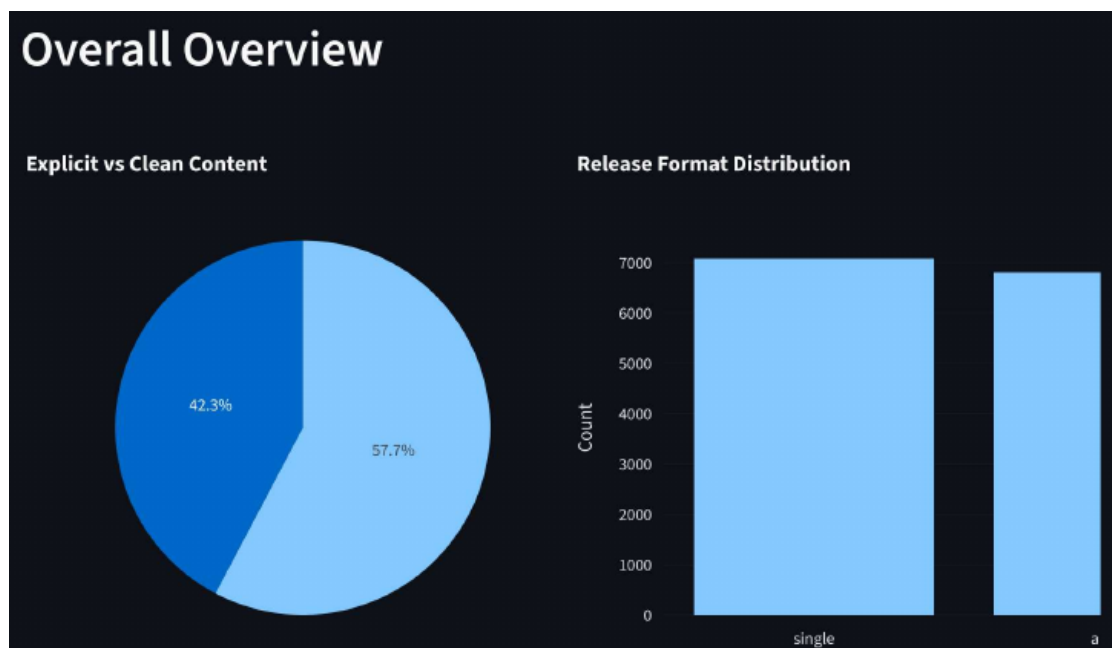


Exhibit B — Dashboard view (Top 25 filter): Explicit vs Clean pie (57.7% / 42.3%) and Release Form Distribution, illustrating the format split within a higher tier. Source: Top_25.pdf, Overview tab.

12.4 Album Structure

Track performance declines monotonically across album-size buckets (Figure 7–9). Small albums (1–10 tracks) average 77.04 popularity at rank 25.64; medium albums (11–15 tracks) average 73.73 at rank 26.45; large albums (16–20 tracks) average 71.97 at rank 26.57; and very large albums (21+ tracks) average 68.62 at rank 29.08. The project-defined Album Size Impact Index (–4.67) compresses the small-versus-large contrast into a single figure. A dilution interpretation is plausible — longer track lists may spread attention more thinly but the design cannot separate album-size effects from confounders such as artist popularity, genre, release strategy, catalog strength, and repeat chart exposure. Album size is therefore discussed as an association throughout.

12.5 Song Duration

Duration is the most structurally stable attribute in the dataset. The mean sits at 3.09 minutes overall and moves by less than four thousandths of a minute between tiers (3.088 Top 10; 3.085 Top 25; 3.091 Top 50). Bucket-level analysis (Figures 10–12) shows the 3–4-minute range carrying the largest volume (12,360 records), the strongest average popularity (77.72), and the best average rank (24.89). The >4-minute group achieves comparable popularity (77.78) but a markedly weaker average rank (29.41), suggesting longer tracks can sustain popularity while facing headwinds in chart positioning. The 2–3-minute bucket dominates volume (13,660 records) with middling performance (75.57 popularity; 25.55 rank). The <2-minute group is small (229 records; 89.96% explicit; 75.09 popularity; 29.97 rank) and should not be overinterpreted.

12.6 Rank-Tier Comparison

Table 4 consolidates the tier profiles. Three gradients are visible: explicit representation rises toward the top (56.27% → 57.70% → 62.44%); average popularity rises toward the top (76.64 → 77.97 → 78.28); and average album size falls toward the top (9.00 → 8.24 → 8.04 tracks). Average duration is flat. The dashboard-tier acceptance values (63.8, 77.0, 84.6 for Top 50, Top 25, Top 10) are mechanically higher for narrower tiers because the composite embeds rank; they describe the tier, not a time trend. Figure 13 summarizes the tier content profile.

Table 4. Top 10 vs Top 25 vs Top 50 content profile

Metric	Top 10 (1–10)	Top 25 (1–25)	Top 50 (1–50)
Explicit share	62.44%	57.70%	56.27%
Clean share	37.56%	42.30%	43.73%
Single share (binary scope)	50.75%	50.99%	47.11%
Album share (binary scope)	49.25%	49.01%	52.89%
Average popularity	78.28	77.97	76.64
Average duration (min)	3.088	3.085	3.091
Average album size (tracks)	8.04	8.24	9.00
Content Acceptance Score (dashboard)	84.6	77.0	63.8

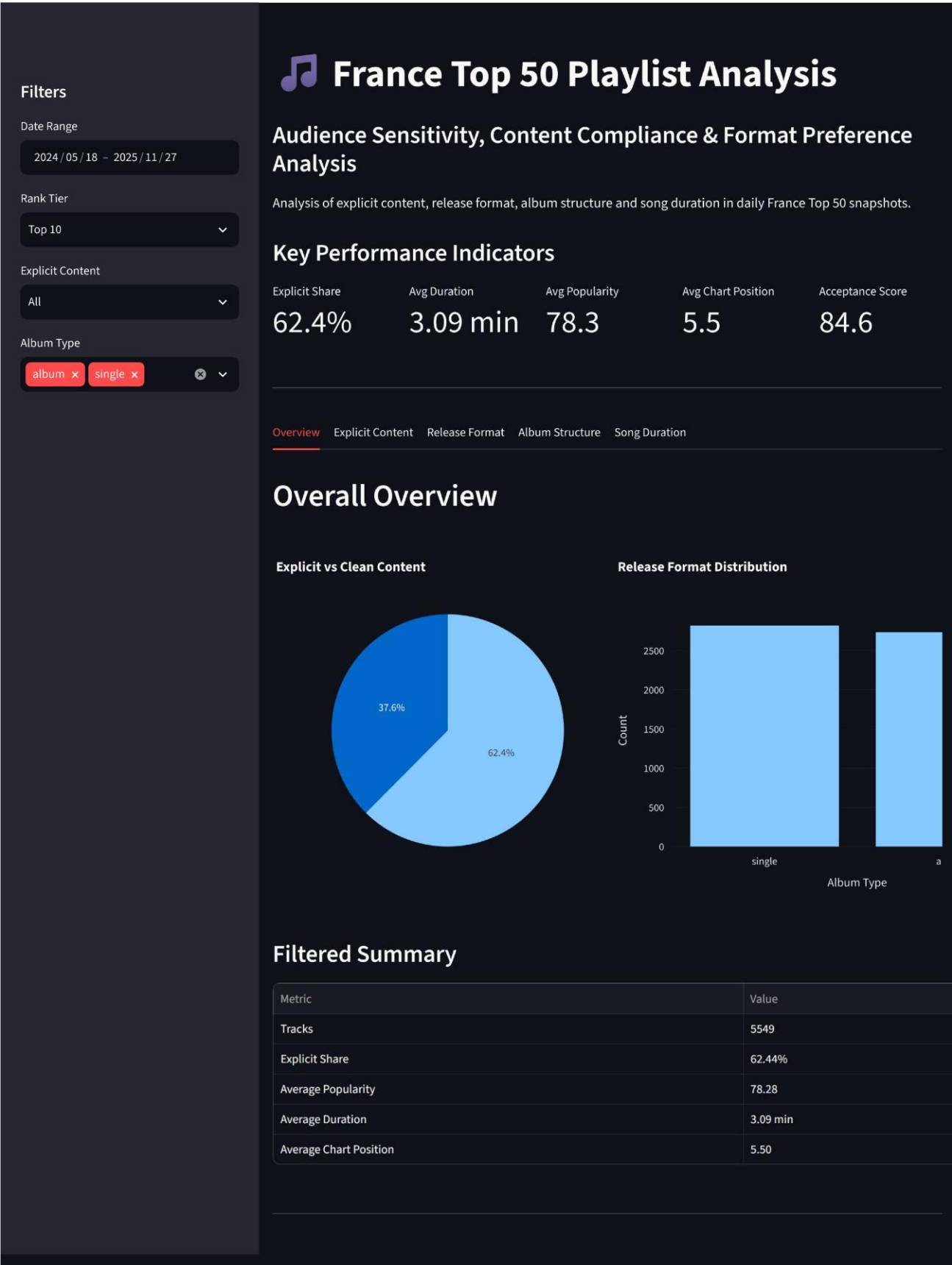
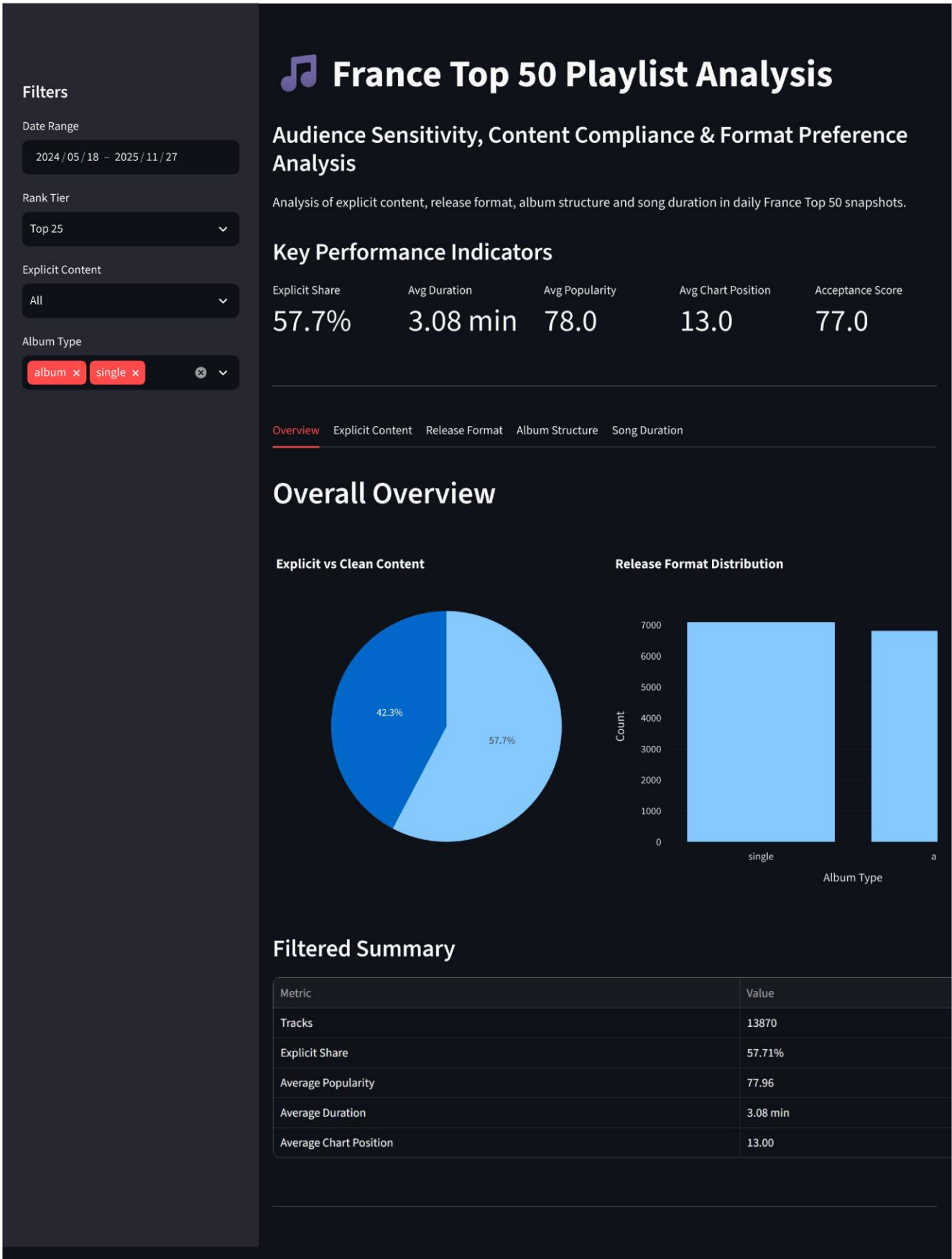
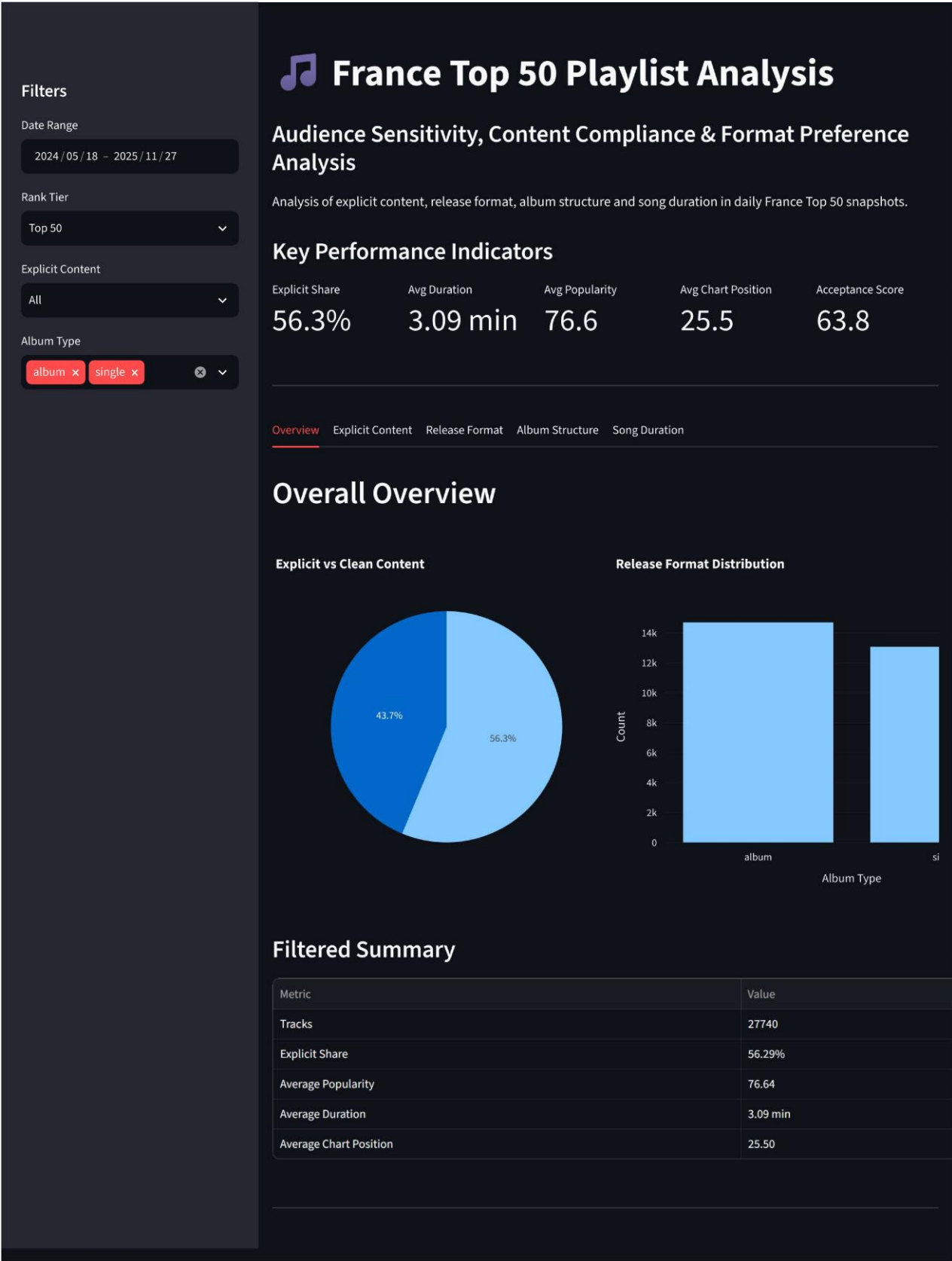


Exhibit C — Streamlit dashboard under the Top 10 rank-tier filter: KPI cards (Explicit Share 62.4%, Avg Duration 3.09 min, Avg Popularity 78.3, Avg Chart Position 5.5, Acceptance Score 84.6), Explicit vs clean pie (62.4% / 37.6%), Release Form Distribution, and Filtered Summary (Tracks 5,549).
Source: Top_10.pdf.





Note: tier values describe nested windows of the same chart. Differences between tiers indicate where attributes are concentrated, not changes over time. The dashboard acceptance values are not directly comparable to the overall 63.82 in Table 3 because the dashboard recomputes the composite within each filtered tier.

The three dashboard exhibits below the same Streamlit dashboard (app.py) under the Top 10, Top 25, and Top 50 rank-tier filters, all with Album Type = album + single corroborate Table 4 directly from the supplied screenshots. Reading the KPI cards left to right across exhibits, Explicit Share climbs from 56.3% (Top 50) to 57.7% (Top 25) to 62.4% (Top 10), Average Popularity rises from 76.6 to 78.0 to 78.3, and the Acceptance Score rises from 63.8 to 77.0 to 84.6, while Average Duration remains 3.09 minutes throughout.

12.7 Monthly Trends

Monthly aggregation (Table 9) reveals that some attributes move materially over the observation window while others barely move. Explicit share ranged from 67.03% (October 2024, the observed maximum) to 45.27% (July 2025, the observed minimum) a swing of more than 21 percentage points. Average popularity ranged from 79.29 (January 2025, maximum) to 72.28 (October 2025, minimum). Average album size ranged from 11.55 tracks (July 2024) to 6.68 tracks (August 2025). Average duration, by contrast, stayed within an approximately 2.98–3.24-minute band across the entire period. Figures 14 and 15 display the explicit-share and popularity monthly series. The practical implication is that content-mix and popularity conditions on the French chart are time-varying and require monitoring, whereas duration norms are comparatively fixed.

Table 9. Monthly trend summary (observed extremes over the study window)

Monthly metric	Maximum (month)	Minimum (month)	Interpretation
Explicit share	67.03% (Oct 2024)	45.27% (Jul 2025)	Content mix varies materially over time
Average popularity	79.29 (Jan 2025)	72.28 (Oct 2025)	Overall chart strength fluctuates
Average duration (min)	≈3.24 (upper end of band)	≈2.98 (lower end of band)	Duration comparatively stable
Average album size (tracks)	11.55 (Jul 2024)	6.68 (Aug 2025)	Album architecture on the chart shifts

13. Statistical Analysis

Welch two-sample t-tests were computed for the three headline group comparisons. All three returned p-values below 0.001, indicating that the observed group differences are large relative to within-group variance under the test assumptions.

Table 10. Exploratory statistical test results (Welch t-tests)

Comparison	Outcome variable	Result	Interpretation
Explicit vs Clean	Popularity	p < 0.001	Observed group difference is statistically significant (exploratory)
Explicit vs Clean	Chart position	p < 0.001	Observed group difference is statistically significant (exploratory)

Single vs Album	Popularity	$p < 0.001$	Observed group difference is statistically significant (exploratory)
-----------------	------------	-------------	--

These results must be read with three qualifications. First, the unit of observation is the track-date record; the same songs recur across the 555 snapshots, so observations within groups are correlated and the effective sample size is smaller than the raw record count. The tests therefore overstate precision, and the p-values are best understood as confirming that the observed differences are not plausibly attributable to chance alone under an independence assumption that is known to be imperfect. Second, statistical significance is not practical significance: with very large record counts, even small differences achieve low p-values, so effect magnitudes (reported in Sections 12 and 14) carry the substantive weight. Third, and most fundamentally, observational group differences cannot be assigned a causal interpretation: explicit status, format, and album size are entangled with artist, genre, and campaign strategy. The tests are therefore presented as exploratory significance testing of observed group differences and are not used as causal evidence anywhere in this paper.

14. Content Acceptance Score Analysis

The project-defined Content Acceptance Score ($0.5 \times \text{Rank Score} + 0.5 \times \text{Popularity}$) provides a single composite lens on chart performance. Overall, the score averages 63.82. Decomposed by content status, clean tracks average 65.295179 (≈ 65.30) and explicit tracks average 62.672826 (≈ 62.67), a clean advantage of approximately 2.62 points on the 0–100 composite.

Table 5. Explicit vs Clean performance profile

Metric	Clean tracks	Explicit tracks
Overall record share	43.73%	56.27%
Top 10 share	37.56%	62.44%
Top 25 share	42.30%	57.70%
Content Acceptance Score (overall)	65.30	62.67
Welch t-test (popularity, position)	—	both comparisons $p < 0.001$ (exploratory)

This pairing a higher average composite for clean tracks alongside a higher top-tier share for explicit tracks is the central nuance of the study and is resolved not by choosing one fact over the other but by recognizing that they describe different questions. The composite averages across all positions and therefore reflects full-chart performance, where explicit tracks are numerous but, on average, positioned and scored slightly lower. The tier share describes the very top of the chart, where explicit tracks are over-represented relative to their full-chart presence. Both statements are true simultaneously: explicit content dominates the chart numerically and concentrates at the summit, while the average explicit record performs slightly below the average clean record on the composite. Section 16 develops the strategic reading of this duality.

15. Key Findings

F1. Explicit content is highly represented on the France Top 50 (56.27% of analytical records) and becomes more concentrated toward the highest ranks (57.70% in the Top 25; 62.44% in the Top 10).

F2. Clean tracks record a higher average project-defined Content Acceptance Score (65.30 versus 62.67), and exploratory Welch t-tests show statistically significant explicit-clean differences in both popularity and chart position ($p < 0.001$, exploratory). The data therefore does not support a simplistic clean-only reading of the market, nor an explicit-superiority claim; it supports a dual-pattern reading.

F3. The full Top 50 is slightly album-heavy (album tracks 52.89% vs singles 47.11%; Single vs Album Ratio 0.89), while singles are marginally more represented at the top (50.75% Top 10; 50.99% Top 25). Album tracks have greater representation across the full Top 50, while singles are slightly more concentrated in higher ranks.

F4. Duration is highly stable overall (mean 3.09 minutes; tier means within ± 0.003 minutes) and across months (approximately 2.98–3.24 minutes). The 3–4-minute bucket shows the strongest average chart position (24.89) and strong popularity (77.72); the >4-minute group matches its popularity (77.78) but ranks materially worse on average (29.41); the <2-minute group (229 records) is too small for firm conclusions.

F5. Smaller album track lists are associated with stronger track performance: average popularity falls monotonically from 77.04 (small) to 68.62 (very large), and average rank worsens from 25.64 to 29.08. The project-defined Album Size Impact Index is -4.67 .

F6. Market characteristics change over time: explicit share swung from 67.03% (October 2024) to 45.27% (July 2025); average popularity ranged from 79.29 (January 2025) to 72.28 (October 2025); average album size ranged from 11.55 to 6.68 tracks. Duration alone was stable.

F7. No finding supports a causal claim. All patterns are chart-level associations within a repeated daily snapshot of the France Top 50, subject to the limitations of Section 19.

16. Discussion

16.1 Explicit-Content Strategy

The most consequential business finding is the combination of majority explicit representation (56.27%) with rising explicit concentration at the top (62.44% of the Top 10). Whatever the mechanisms artist mix, genre composition, editorial selection, labeling practice the observed French chart is not a clean-dominated environment. The data does not support a strategy of avoiding explicit releases in France, and it equally does not support the opposite slogan: the higher average composite for clean tracks (65.30 vs 62.67) warns against treating explicit status as an advantage. The defensible reading is that both content types sustain substantial chart representation, and that performance differences between them are modest relative to the volume both achieve.

16.2 Clean and Explicit Version Planning

Given dual representation, version planning should be release-specific rather than market-uniform. A dual-version strategy preparing clean and explicit masters for tracks with playlist potential preserves optionality across playlist contexts that differ in explicit-content suitability. The observed 62.44% explicit Top 10 share indicates that explicit recordings are not structurally excluded from the highest reaches of the French chart, while the clean composite advantage suggests clean versions remain competitive everywhere. Content labeling should be treated as accuracy infrastructure, not merely compliance: the explicit flag in this dataset is the attribute around which suitability decisions are made, and mislabeled content risks both playlist rejection and reputational exposure.

16.3 Playlist Pitching

Pitching should be context-sensitive. For editorial targets that admit explicit content, the observed top-tier explicit concentration indicates that explicit recordings are a normal, majority presence in high French chart positions. For targets requiring clean content, the strong clean composite supports pitching clean versions without expecting a structural penalty. The Content Acceptance Score, used cautiously as a project-defined screening indicator rather than an industry-standard measure, can help rank candidate tracks for pitch sequencing.

16.4 Release Format Planning

The format pattern album-heavy overall, single-leaning at the top maps naturally onto distinct strategic roles. Singles are the appropriate spearhead for top-tier positioning: they are slightly over-represented in the Top 10 and Top 25. Album tracks, meanwhile, sustain the breadth of the chart (52.89% of the full Top 50), consistent with album campaigns generating multiple charting tracks over time. The data does not license claims that "France prefers albums" or "France prefers singles"; it indicates that album tracks have greater representation across the full Top 50, while singles are slightly more concentrated in higher ranks.

16.5 Album Track-list Strategy

The monotonic album-size gradient (77.04 → 68.62 average popularity across buckets; Album Size Impact Index -4.67) is consistent with an attention-dilution account in which longer track lists are associated with weaker average individual-track performance. The study deliberately does not claim that album length causes weaker performance: artist popularity, genre, release strategy, catalog strength, and repeat chart exposure are all plausible confounders, and no design feature isolates album size. Prudent reading: focused track lists are associated with stronger per-track outcomes in this dataset, and high-potential-track prioritization within longer albums is a low-risk response regardless of which account is true.

16.6 Song-Length Considerations

The 3–4-minute range is the strongest observed duration zone, combining the best average rank (24.89) with strong popularity (77.72) and the largest bucket volume (12,360 records). The >4-minute group's popularity parity with weaker rank (77.78 vs 29.41) suggests that longer tracks can sustain popularity while facing chart-position headwinds an asymmetric pattern compatible with established-hit dynamics rather than debut positioning. The <2-minute group is too small (229 records) to inform strategy. Because duration is stable across tiers and months, duration strategy is a structural constant of the French chart, not a moving target.

16.7 France-Specific Localization

Taken together, the profile that emerges for stronger France Top 50 representation is: explicit-majority but clean-competitive content; single-led top-tier entries within an album-sustained chart; 3–4-minute durations; and tracks from more focused album track lists. This profile is specific to the observed French chart and should not be assumed to transfer to other territories without local validation; equally, global heuristics should not be assumed to hold in France. Localization here means calibrating version planning, pitch sequencing, and track list architecture to this observed profile.

16.8 Monitoring of Changes Over Time

The monthly swings in explicit share (67.03% → 45.27%), popularity (79.29 → 72.28), and album size (11.55 → 6.68 tracks) demonstrate that the French chart’s content profile is a moving object. A strategy calibrated on a single snapshot or on the full-period averages reported here — will drift out of alignment. Continuous monthly monitoring of the KPI set in Section 11 is therefore integral to the strategy, not an optional refinement.

17. Strategic Implications for Atlantic Recording Corporation

The findings aggregate into a four-pillar framework, the France Top 50 Content Strategy Framework, which organizes the implications for release planning, pitching, and compliance.

Table 11. France Top 50 Content Strategy Framework — strategic implications by pillar

Pillar	Observed evidence base	Strategic implication
Pillar 1: Content Sensitivity	Explicit share 56.27% overall, 62.44% Top 10; clean composite 65.30 vs explicit 62.67	Maintain explicit and clean version planning; treat labeling as compliance infrastructure; match version to playlist suitability rather than defaulting to one content type
Pillar 2: Release Format	Albums 52.89% of Top 50; singles 50.75% of Top 10; Single/Album Ratio 0.89	Use singles as top-tier spearheads while leveraging album campaigns for chart breadth; sequence releases to exploit both patterns
Pillar 3: Structural Optimization	3–4 min bucket strongest rank (24.89); album-size gradient with Impact Index –4.67	Treat 3–4 minutes as the reference duration zone; favor focused tracklists and explicit prioritization of high-potential tracks within longer albums
Pillar 4: Continuous Monitoring	Monthly explicit share range 45.27%–67.03%; popularity range 72.28–79.29	Track monthly explicit share, popularity, rank-tier shifts, and content-profile changes; recalibrate strategy on a defined review cycle

Each implication is bounded by the same caveat: the evidence establishes observed association within the France Top 50 chart representation, not causal effect or universal listener preference. The framework is therefore a decision-support structure for evidence-based localization, not a rulebook.

18. Recommendations

The following recommendations follow directly from the findings and use evidence-based language throughout.

- R1.** Maintain both clean and explicit versions of tracks with France playlist potential. The analysis suggests that both content types sustain substantial chart representation, and the observed top-tier explicit concentration indicates that explicit releases face no structural exclusion from high French chart positions.
- R2.** Do not adopt a clean-only France strategy. The data indicates that explicit content represents 56.27% of the chart and 62.44% of the Top 10; a clean-only posture would misalign with observed chart structure. Equally, do not treat explicit status as a performance advantage, given the clean composite advantage (65.30 vs 62.67).
- R3.** Prioritize high-potential singles for top-tier pitching while continuing to service album tracks across the wider chart. The observed pattern supports a spearhead-and-breadth model: singles are slightly more concentrated in higher ranks, while album tracks carry greater representation across the full Top 50.

R4. Avoid assuming that larger album track lists automatically generate stronger individual-track performance. The analysis suggests an association between smaller track lists and stronger per-track outcomes (Impact Index -4.67); where long albums are artistically or commercially necessary, the data supports prioritizing a focused set of high-potential tracks rather than treating all tracks as equal pitch candidates.

R5. Treat 3–4 minutes as the strongest observed duration range for chart positioning. The data indicates that this bucket combines the best average rank with strong popularity. This is a reference point, not a prescription: the >4-minute group demonstrates that longer tracks can sustain high popularity, and duration norms may differ by genre and artist strategy.

R6. Build a France-specific content profile into playlist-pitching materials. Pitching decks and editor communications may appropriately reference the observed French chart profile — explicit-majority representation, 3–4-minute durations, focused track lists as market-context evidence, clearly framed as chart observation rather than listener-behavior measurement.

R7. Monitor the content mix monthly. The observed material variation in explicit share (45.27%–67.03%) and popularity (72.28–79.29) indicates that point-in-time strategy calibration decays; a monthly KPI review against the Section 11 framework may be appropriate, with defined thresholds for strategic reassessment.

R8. Reassess the strategy periodically as the market evolves. Because the observation window ends on 27 November 2025, the findings characterize that window; the observed monthly variability implies that any framework built on it should carry an explicit review-by date and refresh trigger.

R9. Strengthen the evidence base before high-stakes causal decisions. Where Atlantic contemplates decisions that hinge on causal claims for example, that album length itself depresses track performance the current dataset is insufficient; A/B release structures, territory-level streaming telemetry, or listener-behavior data would be required

19. Limitations

L1. Observational dataset. All findings are associations within a chart snapshot; no causal inference is possible, and none is claimed.

L2. No individual listener behavior. The dataset contains no listeners, streams by user, skip rates, completion rates, survey responses, sentiment, or demographics. Statements about "preference" in this paper refer strictly to chart representation and observed association; the analysis does not measure the preferences of all French listeners.

L3. Repeated daily observations. The same songs recur across the 555 snapshots. All record counts overstate the number of independent observations; group comparisons and the exploratory Welch t-tests in Section 13 are affected, and the tests are presented as exploratory significance testing of observed group differences only.

L4. API-derived popularity score. Popularity is a platform API metric whose construction is not controlled by this study; it is treated as a given performance indicator rather than a validated measure of listener demand.

L5. Chart position is not identical to listener preference. Editorial selection, release scheduling, catalog effects, and playlist mechanics all intervene between listener behavior and chart position.

L6. Compilation records. Only nine compilation records exist in the dataset; any statement about compilations would be statistically meaningless, and the binary format analysis excludes them. Their negligible volume does not materially affect overall shares but is disclosed.

L7. Cleaning interventions. The 1 March 2025 duplicated snapshot required retention of one 50-row snapshot, and one incomplete record was removed, leaving one date with 49 records. The cleaned dataset is not perfectly complete, though the residual imperfection is minimal.

L8. Album-concentration metric. The supplied album_concentration.csv produces concentration-rate values above 1 (in excess of 100%) because repeated chart observations are divided by total album size. The metric therefore cannot be interpreted as a true album-level concentration percentage. It is deliberately not used as a headline finding in this paper; it is noted here as an exploratory chart-exposure index at most, with its methodological weakness disclosed rather than concealed.

L9. No genre controls. Genre is absent from the dataset, yet genre plausibly confounds every content attribute examined — explicit share, duration, and album size all vary by genre in the wider industry.

L10. No artist fixed effects. Artist popularity and campaign strategy confound format and album-size comparisons; no artist-level modeling was possible with the available fields.

L11. No country-level comparative baseline. The study profiles France only. Whether the observed patterns are uniquely French or common across markets cannot be assessed without parallel data for other territories.

L12. No streaming volume or retention metrics. Popularity and position are the only performance measures; commercial outcomes (streams, revenue, retention) are not observed.

L13. Time-window bounds. Findings characterize 18 May 2024 – 27 November 2025. The observed monthly variability implies that the profile is time-sensitive and should be periodically refreshed (see R7–R8).

20. Conclusion

This study set out to profile the content structure of the France Top 50 playlist and to translate that profile into strategy for Atlantic Recording Corporation. The research questions can now be answered directly from the evidence.

RQ1 Explicit representation: explicit content accounts for 56.27% of analytical records overall and rises to 57.70% in the Top 25 and 62.44% in the Top 10. Explicit content is highly represented and becomes more concentrated toward the highest ranks.

RQ2 Explicit vs clean performance: clean tracks record the higher average project-defined Content Acceptance Score (65.30 versus 62.67), and exploratory Welch t-tests find statistically significant differences in popularity and position ($p < 0.001$, exploratory). The two facts together clean composite advantage and explicit top-tier concentration define a dual-pattern market in which neither a clean-only nor an explicit-only strategy is supported.

RQ3 Format distribution: album tracks carry greater representation across the full Top 50 (52.89% vs 47.11%), while singles are slightly more concentrated in higher ranks (50.75% of the Top 10; 50.99% of the Top 25).

RQ4 Album size: smaller track lists are associated with stronger performance at every step popularity falls from 77.04 (small) to 68.62 (very large) and rank worsens from 25.64 to 29.08 yielding a project-defined Album Size Impact Index of -4.67 , an association that does not establish causation.

RQ5 Duration: the 3–4-minute range is most strongly represented among higher-ranking tracks (average rank 24.89; popularity 77.72), within an overall duration structure that is exceptionally stable (3.09 minutes mean; approximately 2.98–3.24 monthly band).

RQ6 Composite profile: stronger France Top 50 representation is associated with explicit-majority-but-clean-competitive content, single-led top-tier entries on an album-sustained chart, 3–4-minute durations, and tracks from more focused album track lists.

RQ7 Strategic implications: the findings support the four-pillar France Top 50 Content Strategy Framework — content sensitivity, release format, structural optimization, and continuous monitoring as a basis for France-specific, evidence-based localization rather than any blanket clean/explicit or single/album rule.

The central evidential thread of the paper is worth restating: the observed French chart is explicit-majority and explicit-concentrated at the top, yet clean-performing on average; album-sustained overall, yet single-leaning at the summit; durationally stable; and favorable to focused track lists. Because all of these are chart-level associations within a repeated daily snapshot, their proper strategic use is calibration and monitoring not causal rule-making. Used with that discipline, the analysis provides Atlantic Recording Corporation with a defensible, France-specific evidence base for content, format, and localization decisions, together with a monitoring framework to keep that evidence current as the market evolves.

References

Internal and project sources:

Atlantic Recording Corporation project documentation (2026). Project brief and business requirements: “Audience Sensitivity, Content Compliance & Format Preference Analysis of France Top 50 Playlist.” Supplied project materials: cleaned_france_top50.csv; kpi_summary.csv; rank_tier_content_profile.csv; format_by_rank_tier.csv; duration_analysis.csv; album_size_analysis.csv; monthly_analysis.csv; album_concentration.csv; app.py; Top_10.pdf; Top_25.pdf; Top_50.pdf.

Methodological sources:

Welch, B. L. (1947). The generalization of Student’s problem when several different population variances are involved. Biometrika, 34(1–2), 28–35.

Spotify for Developers (2024–2025). Web API reference: Track object (popularity field; explicit flag; album total_tracks). Spotify AB. Accessed via the project data pipeline.

Note: no further external sources were used; no references have been fabricated for this study.

Appendix

Appendix A. KPI formulas (exact definitions)

Table A1. KPI formula reference

KPI	Exact formula	Overall value	Status
Explicit Content Share	$\text{explicit_records} / 27,749 \times 100$	56.27%	Standard
Clean Content Share	$\text{clean_records} / 27,749 \times 100$	43.73%	Standard
Clean Dominance Ratio	$\text{clean_records} / \text{explicit_records}$	0.78	Project-defined
Single vs Album Ratio	$\text{single_records} /$	0.89	Standard ratio, binary scope

	album_records (compilations excluded)		
Average Song Duration	mean(duration_min)	3.09 min	Standard
Average Popularity	mean(popularity)	76.64	API-derived
Album Size Impact Index	mean(popularity album > 10 tracks) – mean(popularity album ≤ 10 tracks)	–4.67	Project-defined
Rank Score	$100 \times (51 - \text{position}) / 50$	—	Project-defined component
Content Acceptance Score	$0.5 \times \text{Rank Score} + 0.5 \times \text{popularity}$	63.82 (clean 65.295179; explicit 62.672826)	Project-defined composite

Appendix B. Supporting analytical outputs

B1. Duration buckets. <2 min: 229 records, popularity 75.09, rank 29.97, explicit share 89.96%. 2–3 min: 13,660 records, popularity 75.57, rank 25.55, explicit share 56.27%. 3–4 min: 12,360 records, popularity 77.72, rank 24.89, explicit share 53.24%. >4 min: 1,500 records, popularity 77.78, rank 29.41, explicit share 76.13%.

Table 7. Duration bucket performance

Duration bucket	Track count	Avg popularity	Avg rank	Explicit share
<2 min	229	75.09	29.97	89.96%
2–3 min	13,660	75.57	25.55	56.27%
3–4 min	12,360	77.72	24.89	53.24%
>4 min	1,500	77.78	29.41	76.13%

B2. Album-size buckets. Small (1–10): 1,920 tracks, popularity 77.04, rank 25.64. Medium (11–15): 6,621 tracks, popularity 73.73, rank 26.45. Large (16–20): 4,180 tracks, popularity 71.97, rank 26.57. Very Large (21+): 1,951 tracks, popularity 68.62, rank 29.08.

Table 8. Album size performance

Album size bucket	Tracks	Avg popularity	Avg rank
Small (1–10)	1,920	77.04	25.64
Medium (11–15)	6,621	73.73	26.45
Large (16–20)	4,180	71.97	26.57
Very Large (21+)	1,951	68.62	29.08

Table 6. Single vs Album performance by rank tier (binary scope, compilations excluded)

Tier	Single share	Album share
Top 10	50.75%	49.25%
Top 25	50.99%	49.01%
Top 50	47.11%	52.89%

B3. Statistical tests. Welch t-tests (exploratory): explicit vs clean popularity, $p < 0.001$; explicit vs clean chart position, $p < 0.001$; single vs album popularity, $p < 0.001$. Repeated daily observations violate independence assumptions; results are not confirmatory.

B4. Album-concentration metric caveat. The album_concentration.csv concentration-rate values can exceed 1 because repeated chart observations are divided by total album size; the metric is therefore not a valid percentage and is

excluded from headline findings. At most it may be reframed as an exploratory chart-exposure index. This weakness is disclosed in limitation L8.

B5. Dashboard exhibits. The three dashboard screenshots (Top_10.pdf, Top_25.pdf, Top_50.pdf) show the same Streamlit dashboard (app.py) under different rank-tier filters, with Album Type = album + single; compilation records are excluded from displayed KPI values and format charts. The exhibits are reproduced in Section 12 of this paper.

Appendix C. Figure register (recommended exhibits)

Figure 1. Project analytical framework. Purpose: situates validation → feature engineering → KPI computation → tier/trend analysis → strategy. Interpretation: the framework enforces the descriptive/exploratory/interpretive separation of Section 9.

Figure 2. Explicit vs clean distribution. Shown in Exhibit A (Section 12.2): the 56.3% / 43.7% overall pie (15,614 vs 12,135 records).

Figure 3. Explicit content share by rank group. Shown across Exhibits A–C (Sections 12.2, 12.6): the rising gradient 56.3% (Top 50) → 57.7% (Top 25) → 62.4% (Top 10); the key visual evidence that explicit content concentrates at the highest ranks.

Figure 4. Single vs Album distribution. Shown in Exhibit A: the overall 47.11% / 52.89% split under the binary album + single scope.

Figure 5. Single vs Album representation by rank tier. Shown across Exhibits A–C: the crossover — albums lead the full Top 50; singles edge ahead in the Top 10 and Top 25.

Figure 6. Popularity comparison by release format. Visualizes the single-album popularity difference corroborated by the exploratory Welch test ($p < 0.001$).

Figure 7. Album size distribution. Shows the track-count distribution across the four album-size buckets (1,920 / 6,621 / 4,180 / 1,951).

Figure 8. Album size vs popularity. Scatter/binned view of the monotonic negative gradient underlying the –4.67 Impact Index.

Figure 9. Popularity by album-size bucket. Bar view: 77.04 / 73.73 / 71.97 / 68.62.

Figure 10. Song duration distribution. Shows the concentration of the chart around the 3-minute center (mean 3.09).

Figure 11. Popularity by duration bucket. Bar view: 75.09 / 75.57 / 77.72 / 77.78.

Figure 12. Duration vs chart position. Shows the U-shaped rank pattern across buckets (24.89 best at 3–4 min; 29.41 at >4 min; 29.97 at <2 min).

Figure 13. Top 10 vs Top 25 vs Top 50 content profile. Shown directly in Exhibits C–E (Section 12.6): the three dashboard KPI card rows and pies; consolidates explicit share, popularity, duration, album size, and acceptance across tiers (Table 4 values).

Figure 14. Monthly explicit-content trend. Range 45.27%–67.03%; peak October 2024; through July 2025.

Figure 15. Monthly popularity trend. Range 72.28–79.29; peak January 2025; through October 2025.

Figure 16. Correlation matrix. Exploratory bivariate associations among duration, album size, popularity, and position; consistent with, and no stronger than, the group-level patterns reported in Section 12.